VU AI HUB · LECTURE NOTES

Project Collective Intelligence

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Written by Inaaya and shared on Notion. Original page: cheddar-wall-f3b.notion.site/3729801a930980eaa304ea5c7fd21145

2nd of June

The Fields/Domains

Collective Intelligent Systems

> *A group of limited, local agents produces group behavior that is robust, adaptive, and qualitatively different from any individual rule*

- Decentralized: no central controller, removing any dingle agent leaves the system functional. No agent knows the global state
- Local: each agent sees only its neighborhood - in space, in time, or on a network. Information is bounded and noisy
- Emergent: macroscopic behavior is qualitatively different from microscopic rules. The group does what no agent **can do by themselves?**

Biological collectives: biology gives us the rules → found, not designed

- Rules are the product of evolution
- Our task: infer them from observation and model them

Engineered collectives: we write the rules → designed, not found

- Rules are chosen
- Our task: **???**

Every system can be specified by three questions:

1. Agent: Who is the agent? What is its internal state? What rule maps [state + input] **????**
2. Environment: In which environment lives the agent? What stimuli are there? Does it have memory?
3. Interaction: Direct signaling, or via the environment?

Examples

Flocking

- Hundreds of bodies moving together in the same direction as if they share the same body and brain

  Reynold's three rules 1986

  1. Separation
     - Don't crash into your neighbor
  2. Alignment
     - Match your neighbors' direction
  3. Cohesion
     - Steer toward the local centroid (the center of the neighbors)

  The Vicsek model 1995

Looked at the interaction radius and noise

- Increasing the interaction radius led to cohesion
- Decreasing the noise led to cohesion

Couzin et al. three zones 2002-2005

Looked at the repulsion, orientation and attraction

Foraging

- Communication by changing the world, not by talking to each other

  Pheromone Coding

  How biological nervous systems detect, decode, and interpret chemical signals to drive instinctive behaviors like mating, alarm, and social hierarchy

Division of labor

- Thousands of jobs. No manager. Stable ratios

  Response Threshold

  A theoretical framework explaining how complex systems—like social insect colonies or decentralized multi-agent algorithms—achieve task specialization without central coordination

  - Temporal Polytheism
    - Honeybees move through tasks with age: nurse → builder → guard → forager
  - Morphological caste
    - Bees: queens, workers or soldiers
    - Distinct physical traits that alter their neural and psychological sensitivity, predisposing them to different tasks based on these evolved response thresholds
  - Behavioral threshold
    - The exact number or proportion of people who must adopt a new behavior before an individual decides to do the same

Decision-making

- A swarm of bees picking a home
  1. Commitment
     - Quorum threshold
       - Scouts at the nest site sense crowding density → commit
  2. Recruit
     - Waggle Run
       - The scout performs a figure-eight dance. The angle to vertical encodes the direction of the site relative to the sun
     - Duration = Quality
       - The number of dance circuits directly encodes site quality. Better sites get longer, more vigorous dances

  The Speed-Accuracy Tradeoff

Synchronization

Kuramoto model

Opinion Dynamics

- Consensus, polarization, echo chambers
  - The collective system you live inside every day

Voter Model

- Mechanism
  - Each agent copies one randomly chosen neighbor per step. No averaging, no memory, pure stochastic replacement
- Who wins?
  - **???**
- Key property
  - Unbiased random walk on opinion space
  - Consensus is inevitable on any finite connected graph

DeGroot Model

- Mechanism
  - Each agent updates its opinion to a weighted average of all neighbors. Weights encode trust or social influence
- Polarization?
  - DeGroot always converges. Polarization only appears with disconnected components or stubborn (non-updating) agents

Bounded Confidence

- Mechanism
  - Each agent averages only over agents within confidence window. Opinions outside of it are ignored
- Polarization
  - Below a critical confidence bound the population splinters into stable clusters. Agents in different clusters stop communicating entirely
- Critical Threshold